Response to the editor and reviewers

Characterisation and Quantification of Data Centre Flexibility for Power System Support

Previous manuscript: APEN-D-25-21553
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Dear Editor and Reviewers,

Thank you for the detailed comments. We have substantially revised the manuscript and provide a point-by-point response below. The revision includes full-year operation and resource sensitivities, an explicit optimisation workflow, transparent workload inputs, component cost accounting, numerical checks, clearer figures and more precise limits on the flexibility and network claims.

In accordance with the decision letter, the cover letter identifies the previous manuscript, and we provide clean and marked revised copies. The marked copy compares the revised manuscript with the version submitted as APEN-D-25-21553. Blue text identifies additions or replacements; deleted material is suppressed. Page numbers below refer to the clean revised manuscript. Figure numbers have changed after insertion of the workflow diagram. Reviewer comments are reproduced in full.

Reviewer 2

Comment 2.1 — Novelty and the role of the UPS model

Reviewer: *The UPS model established in this paper lacks significant differentiated innovation compared with traditional "data centre + energy storage" models. Existing studies have extensively covered topics related to the coordinated optimization of data centres and energy storage, and this model fails to reflect breakthrough technical contributions, making it difficult to highlight the innovative value of the research.*

Response

We agree that the UPS formulation is a standard storage model and should not be presented as a new battery model. We have clarified that the contribution lies in the integrated, duration-aware flexibility assessment. IT arrivals and delay windows, UPS energy, thermal states and TES energy jointly constrain the deviation that the facility can sustain from its cost-minimising baseline. Scenario 3 evaluates start time, requested power deviation and service duration while enforcing comparative recovery conditions.

The revised introduction and literature review distinguish this contribution from asset-level scheduling and earlier coordinated energy-management studies. The discussion explains the cross-asset interactions revealed by the model, while expressly avoiding a claim that the present results measure an integration premium over separately optimised subsystems; that would require an additional comparator. The UPS model remains conventional so that the contribution and its limits are explicit.

Location: Introduction, pp. 1 and 4–5; Sections 2–3, pp. 5–10; Section 4.3, p. 13; Section 5.5, pp. 18–19.

Comment 2.2 — Figure clarity, terminology and presentation

Reviewer: *There are obvious confusions in the paper's language expression and figure presentation: first, the logic of describing figures is unclear. For example, it is mentioned that "Figure 7 shows six stacked bar charts arranged into a 3-by-2 grid... Figure 7 shows negative ΔP (upward flexibility) and Figure 8 shows positive ΔP (downward flexibility)", which is likely to cause confusion for readers. Readers do not know whether this sentence refers to Figure 7 or Figure 8; second, the font size in the figures is too small, inconsistent with the font size in the main text, affecting readability; third, the use of abbreviations is not unified, with "DC" used in some places and "data centre" in others; fourth, the two subfigures in Fig.4 have significant style differences and are not suitable for presentation as subfigures of the same figure.*

Response

We have revised the figure presentation and the accompanying interpretation. The upward-flexibility and downward-flexibility discussions are now separate and identify the sign convention explicitly: negative ΔP denotes a load reduction, while positive ΔP denotes a load increase. The component plots carry panel labels (a)–(d), and their captions identify the start times and requested magnitudes. All plots are now drawn at their printed size, so figure text is 7–8 pt and consistent with the main text; Figures 2 and 3 have been redrawn in the same style. The plots use consistent colours and are supplied in vector form, and the multi-panel component plots span the page width.

The power/price and IT-workload panels retain their complementary roles but now share a common time axis and consistent presentation. To keep terminology uniform, the abbreviation “DC” is no longer used: “data centre” is written in full throughout the text, captions and figures. Inserting the workflow figure changes the numbering: original Fig. 4 corresponds to revised Fig. 5, and the upward/downward component plots are now Figs. 8 and 9.

Location: Figs. 2–3 and 5–10, pp. 7–8 and 14–18; Section 5.2, pp. 14–17; terminology throughout.

Comment 2.3 — Case-study structure and additional sensitivities

Reviewer: *The case study section is verbose, with an excessive proportion of scenario descriptions, while only the operational results of three scenarios are actually compared, making it difficult to fully reflect the contributions of this research. It is recommended to supplement multi-dimensional case analyses, such as the model response comparison under different electricity price fluctuation ranges, IT load flexibility ratios, and equipment parameters to enhance the universality and persuasiveness of the research.*

Response

We have reorganised the case-study explanation around the three-scenario workflow and substantially expanded the numerical evidence. The annual comparison now uses linked daily optimisation over all of 2025. Fourteen completed annual chains cover the reference case, the central coordinated case and one-at-a-time changes in flexible workload share, UPS energy capacity and TES energy capacity at multipliers 0.5, 0.75, 1.25 and 1.5. The flexible-share construction preserves total arriving work and clips the flexible portion at the total workload.

The full year includes the observed variation in signed reference prices, including negative prices. We have not added a controlled price-volatility sweep: annual exposure to realised prices does not isolate the causal effect of volatility. The revised text states this limit explicitly, so the sensitivity results support conclusions about the tested workload and equipment assumptions, not universality across price regimes or sites. The full endpoint table and solver exceptions are supplied in the supplement.

Location: Section 4 and Fig. 4, pp. 10–13; Sections 4.4 and 5.4, pp. 13 and 17–18; Fig. 10, p. 18; Supplementary Sections S4–S5.

Comment 2.4 — Cost composition and the reported saving

Reviewer: *The paper only mentions a "10% cost reduction" in the conclusion without providing specific cost composition and comparative data to support this claim. It is suggested to add a cost comparison table in the case*

study section, clearly listing the sub-item costs (e.g., IT load, cooling system, energy storage operation) and total costs of each scenario, so as to improve the credibility of the conclusion.

Response

We have added a full annual signed electricity-cost breakdown. The central annual reference cost is GBP 550,503.59 and the coordinated cost is GBP 520,876.68, giving GBP 29,626.91 (5.38%) savings. IT electricity expenditure falls by GBP 16,722.17 and direct CRAC expenditure by GBP 14,425.46; TES charging adds GBP 6,897.29, net UPS exchange reduces expenditure by GBP 5,376.57, and auxiliary expenditure is unchanged. Table VI reconciles these terms.

The revised annual result replaces the 10.02% saving previously reported for a single day. The annual 5.38% covers a different assessment period and should not be interpreted as a direct percentage-point comparison with the single-day result. These components are accounting terms within a coordinated dispatch, not independent asset profits. The text also reports the increase in annual electricity use and peak import and excludes service revenues and degradation costs from the valuation.

Location: Section 5.3, p. 17; Table VI, p. 18; Conclusion, p. 19; Supplementary Section S4.

Comment 2.5 — Problem class, implementation and reproducibility

Reviewer: *The core information of the optimization problem is not clearly elaborated. For example, the problem type, the adopted solution algorithm are not specified, resulting in insufficient reproducibility of the research method. It is recommended to supplement relevant technical details.*

Response

The revised manuscript identifies the solved model as a mixed-integer linear programme implemented in Pyomo with HiGHS. Binary variables enforce storage modes and the IT power curve uses a four-segment piecewise-linear DLOG representation with non-uniform breakpoints. The manuscript specifies this approximation and its maximum interpolation error (6.241 kW, or 0.624% of the 1 MW IT rating), the optimisation objectives, temporal boundaries, state/workload carry-over and event recovery criteria.

The supplement records the retained software versions, hardware, time limits, requested and accepted gaps, and physical/accounting checks. It distinguishes feasible time-limited incumbents from proven optimal solutions. Three annual sensitivity horizons exceed the normal 1% acceptance threshold and are disclosed with their small absolute daily bound widths; these are not hidden by a blanket optimality claim. All 148 reported positive-duration event dispatches meet the separate event gap criterion. The adopted inputs and parameters are reported in full in the manuscript and supplement, and the electricity prices are publicly available.

Location: Sections 4.2–4.3, pp. 12–13; Fig. 4, p. 11; Supplementary Sections S2–S5; Data availability, p. 19.

Comment 2.6 — Grid connection and network implications

Reviewer: *According to the optimization strategy proposed in this paper, the data centre load will show a significant peak-valley difference (as shown in Fig.4). However, the paper fails to fully demonstrate the feasibility and adaptability of the data centre model connecting to the power system under this load characteristic, and lacks analysis on the impact on power grid stability and corresponding solutions.*

Response

We agree that cost-minimising operation can increase the network peak and that this affects the interpretation of flexibility. The revised results explicitly report an increase in annual peak import from 974.9 to 1,682.5 kW and annual grid electricity from 6.718 to 6.789 GWh, despite the lower reference-price expenditure.

The model assesses feasible facility responses at the connection point; it does not contain an upstream power-flow or dynamic-stability model. We therefore do not claim that the optimised load profile is

automatically admissible on a particular network or improves stability. The discussion identifies topology, voltage and thermal limits and other concurrent loads as necessary inputs for a subsequent network assessment. The reported responses are candidate flexibility offers under the stated facility assumptions, to be screened against those network constraints and service requirements.

Location: Section 5.3, p. 17; Section 5.5, pp. 18–19; Conclusion, p. 19.

Comment 2.7 — Workload data sources and representativeness

Reviewer: *Fig.2 presents the IT load ratio data used in this study, but does not specify the data source in detail. The reliability and representativeness of the data are not supported. It is recommended to supplement a clear source description and data verification basis.*

Response

We have made the status and construction of the workload inputs explicit. They are stylised adopted profiles informed by the cited workload literature, rather than measured site data, a statistically representative facility sample or a deterministic reconstruction of a published trace. Table II summarises the evidence informing the assumptions, while Tables III–IV and Supplementary Table S1 expose the numerical inputs and their interpretation.

The model uses the 96 quarter-hour utilisation values directly. The main-paper workload plot/table displays hourly arithmetic means, not the first quarter-hour in each hour. The 24 hourly tranche-share rows are expanded by repeating each row for its four consecutive quarter-hours. The exact flexible/inflexible arrivals and tranche shares are tabulated in Supplementary Table S1, and the supplement explains the delay-window convention and clipping rule used in the sensitivity analysis. We have not represented a network electricity-demand dataset as measured CPU utilisation. The revised limitations and sensitivity results qualify the transferability of these adopted profiles.

Location: Section 3.1, pp. 6–8; Tables II–IV and Fig. 2; Section 5.4 and Section 5.5 limitations; Supplementary Section S1 and Table S1.

Reviewer 5

Comment 5.1 — Aggregate modelling rationale and distinct contribution

Reviewer: *While co-scheduling IT workloads, UPS batteries, and cooling infrastructure provides an insightful integrated approach, the current presentation fails to clearly distinguish the proposed contribution from existing studies on the coupled operation of data center subsystems. The framework appears more like a combination of three simplified subsystem models, while the underlying rationale behind these simplifications is not sufficiently explained. The authors should clarify how the proposed modeling approach balances computational efficiency with the necessity of preserving the key operational and physical variables that determine data center flexibility provision.*

Response

We have clarified both the modelling purpose and the variables retained. The facility-level model preserves arriving computational work and eligible execution windows, UPS and TES energy, storage power limits and mode exclusivity, and the thermal states that couple IT activity to cooling demand. These are the quantities that constrain a grid-facing power deviation and its duration. Aggregation avoids modelling individual instructions, rack airflow and task dependencies, making repeated horizon and duration evaluations tractable.

This abstraction does not establish task-level quality of service for arbitrary applications. The text now explains that CPU-hours are a processor-occupancy representation and states the limitations of the adopted workloads, fixed environmental assumptions and quarter-hour resolution. The duration assessment uses a common operating baseline and explicit recovery conditions across the coupled resources. The integrated-dispatch discussion illustrates those mechanisms without claiming a measured integration premium against a separately optimised comparator.

Location: Introduction and Sections 2–3, pp. 1 and 4–10; Sections 4.2–4.3, pp. 12–13; Section 5.5, pp. 18–19; Supplementary Sections S1–S3.

Comment 5.2 — Introduction structure and research focus

Reviewer: *The introduction could benefit from a more structured and direct focus on data centers. The current narrative transitions from general energy demand to data center consumption, and then to general flexibility needs and data center flexibility resources and definition. This structure makes the research motivation less direct.*

Response

The revised introduction opens with the expansion and concentrated electricity demand of data centres, connects this directly to power-system flexibility, and sets out the facility resources relevant to the study. The research gap and contributions distinguish coordinated energy scheduling from quantifying feasible power-deviation durations relative to a common operating baseline. The revised abstract likewise states the integrated framework, annual baseline, duration assessment and principal findings directly.

Location: Abstract and Introduction, pp. 1 and 4–5; Section 2, pp. 5–6.

Comment 5.3 — Transparency of workload profiles and ratios

Reviewer: *The workload utilization profiles and flexible/inflexible workload ratios are central to the results, but their data sources and construction process require greater transparency. The authors should provide detailed underlying data sources and the precise derivation steps used to establish these baseline workload characteristics.*

Response

We have expanded the workload provenance and construction description, as also detailed in our response to Reviewer 2, comment 7. The profiles are explicitly described as stylised assumptions informed by literature, with no claim that the chosen flexible/inflexible ratio is universal or directly measured. Table II identifies the supporting literature and its role; Tables III–IV and Supplementary Table S1 give the adopted quantities.

The supplement specifies the 96 quarter-hour arrays, the arithmetic averaging used for hourly presentation, and the repetition of each hourly tranche row across four consecutive intervals. It explains arrival/tranche work, final eligible execution intervals and the sensitivity clipping rule that preserves total work. The complete quarter-hour inputs are tabulated. These changes make the calculation reproducible while separating the supporting literature from the authors' numerical assumptions.

Location: Section 3.1, pp. 6–8; Tables II–IV; Supplementary Section S1 and Table S1.

Comment 5.4 — Justification of utilisation multiplied by duration

Reviewer: *The simplification of computational demand as utilization multiplied by duration is acceptable for an aggregate macro-level model, but the authors should provide relevant literature references or brief discussion to justify the physical and operational validity of this simplification.*

Response

We have added a discussion with references to Dayarathna et al., the Google cluster-trace analysis of Rasheduzzaman et al. and Cao et al. The product of CPU utilisation and execution duration is interpreted as accumulated processor occupancy at the aggregate facility level. For example, 50% utilisation for two hours and 100% for one hour represent equal CPU-hour demand in this abstraction.

This equivalence is not a claim that heterogeneous real jobs have identical instruction throughput or can always trade utilisation for runtime without other constraints. The manuscript limits its scope to aggregate CPU-work scheduling, retains eligible execution windows and conservation of work, and identifies task dependencies and GPU-specific controls as outside the representation.

Location: Section 3.1.1 and Eq. (2), p. 8; Section 3.1.2, pp. 8–9; Section 5.5 limitations, p. 19.

Comment 5.5 — Optimisation workflow

Reviewer: *The equations are very detailed, but the overall optimization workflow is not sufficiently intuitive. The authors should provide a more logically organized explanation, or preferably a workflow diagram, to clarify the inputs, outputs, objective functions, decision criteria, and information flow of each scenario.*

Response

We have added a three-scenario workflow diagram and clarified the inputs, objectives and information flow. Scenario 1 establishes a workload-at-arrival reference with storage disabled; cooling is still determined consistently with the facility constraints. Scenario 2 co-optimises operation against signed reference prices over linked daily horizons, committing each day and passing physical states and outstanding work to the next.

Scenario 3 starts from the resulting undisturbed trajectory. For each requested start time and power deviation, the duration search solves event-plus-recovery dispatch problems and applies the recorded feasibility and solution-quality checks. The output is a grid of certified feasible durations and corresponding component dispatches. The revised explanation distinguishes timeouts from infeasibility and notes that assessment caps and a moving recovery reference prevent interpreting every reported duration as a proven global maximum.

Location: Fig. 4, p. 11; Sections 4.1–4.3, pp. 11–13; Supplementary Sections S2–S3.

Comment 5.6 — Selection and meaning of component decompositions

Reviewer: *Although the manuscript states that the Scenario 2 optimization is rerun for each duration in Scenario 3, the selection and interpretation of the reported component-level decomposition remain unclear. The authors should clarify whether Figs. 7 and 8 show just simply one feasible solution returned by the solver? If multiple feasible dispatch solutions exist for the same start time, flexibility magnitude, and duration, the authors should clarify how the component contributions shown in the figures are determined.*

Response

We have clarified that each decomposition is an accepted feasible MILP dispatch for the specified start time, power deviation and certified event duration. The event-plus-recovery signed electricity-cost objective selects among feasible dispatches, subject to the recorded solver gap. It is therefore one economically preferred realisation; the model does not establish a unique physical attribution or average over all feasible optima.

The plotted components are differences from the same undisturbed trajectory and sum to the metered grid-power deviation. Opposing component responses are retained because some assets may absorb energy while others over-deliver the requested net change. Recovery is assessed through terminal storage, temperature and outstanding-work conditions. The figures distinguish the event window from the unplotted recovery period and mark unresolved adjacent-duration trials. The independent component accounting check covers all 148 positive-duration dispatches.

Location: Section 4.3, p. 13; Section 5.2 and Figs. 8–9, pp. 14–17; Supplementary Sections S3 and S5.

Comment 5.7 — Insights from modelling the subsystems together

Reviewer: *The current results mainly describe the contribution of each component. It would be helpful if the authors could further discuss what additional insights can be obtained from modeling these subsystems together rather than separately.*

Response

We have added a dedicated discussion of the value of integrated co-optimisation. It explains the coupled reduction in IT heat and cooling demand when work is deferred, the change from UPS-dominated morning reductions to workload/cooling-dominated afternoon reductions, and the interaction of TES

charging and direct cooling with the other assets. The component plots show simultaneous opposing responses that meet a single connection-point target.

These mechanisms demonstrate why installed capacity alone does not determine feasible duration: the baseline workload, storage energy and thermal state matter together. For example, many 100–200 kW reductions end at 07:00 or 18:00 because the undisturbed schedule already supplies the whole IT load from the UPS and has deferred all eligible work at those times. We also state the evidential limit explicitly. The present study reveals coupled dispatch mechanisms but does not quantify an integration premium over independently optimised subsystems, because that comparator was not run.

Location: Section 5.2 and Figs. 8–9, pp. 14–17; Section 5.5, pp. 18–19.

Comment 5.8 — Distinguishing charging and discharging in the resource plot

Reviewer: *In Fig. 5, same colors are used to denote different positive and negative components. Different hatch patterns or separate legends would improve readability.*

Response

We have revised the resource-power plot to distinguish positive consumption/charging from negative UPS discharge. UPS discharge now uses a hatch pattern and a separate legend entry from charging. The caption explains the sign convention and clarifies that TES discharge reduces chiller electricity demand rather than injecting electricity. The plot is supplied as a full-width vector figure for readability. The insertion of the workflow diagram means the original Fig. 5 is now Fig. 6.

Location: Revised Fig. 6 and its caption, p. 15.